Direct Memory Access

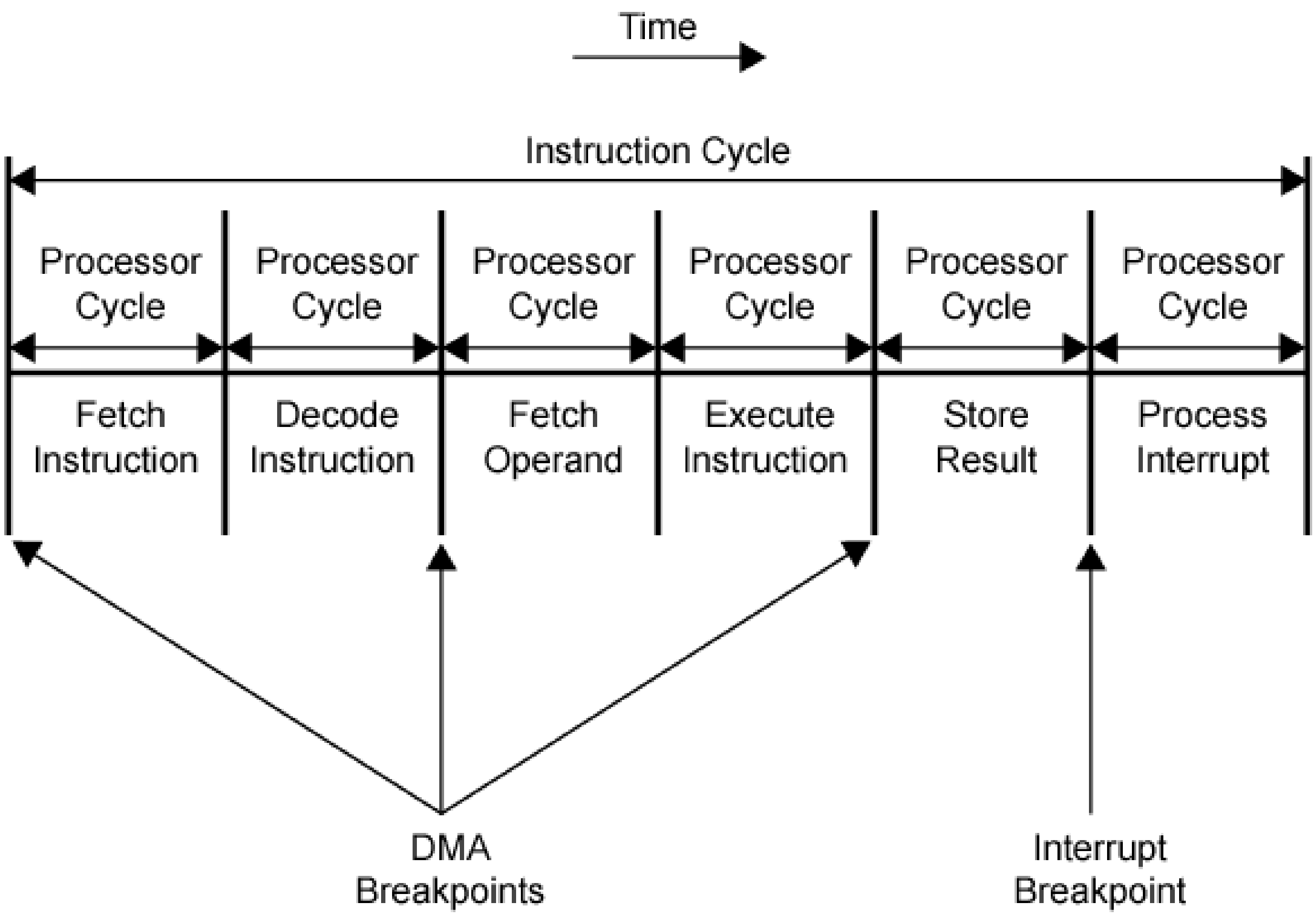
8237

- DMA or Direct Memory Access Controller is an external device that controls the transfer of data between I/O device and memory without the involvement of the processor. It holds the ability to directly access the main memory for read or write operation.
- DMA controller was designed by **Intel**, to have the fastest data transfer rate with less processor utilization.

- The microprocessor first fetches the instruction and then decodes it, then further execute it. But individually if the processor is performing all the task inside the system then it unnecessarily keeps the processor busy all the time.
- To enhance the performance of the processor, an external device is used that can manage data transfer operation between peripherals and memory with least CPU utilization.

- It is nothing but hardware controlled data transfer, where the address and control signals required for transferring the data is generated by an external device.
- This method of data transfer is known as direct memory access and the external device used for this purpose is known as DMA controller.

DMA and Interrupt Breakpoints During an Instruction Cycle



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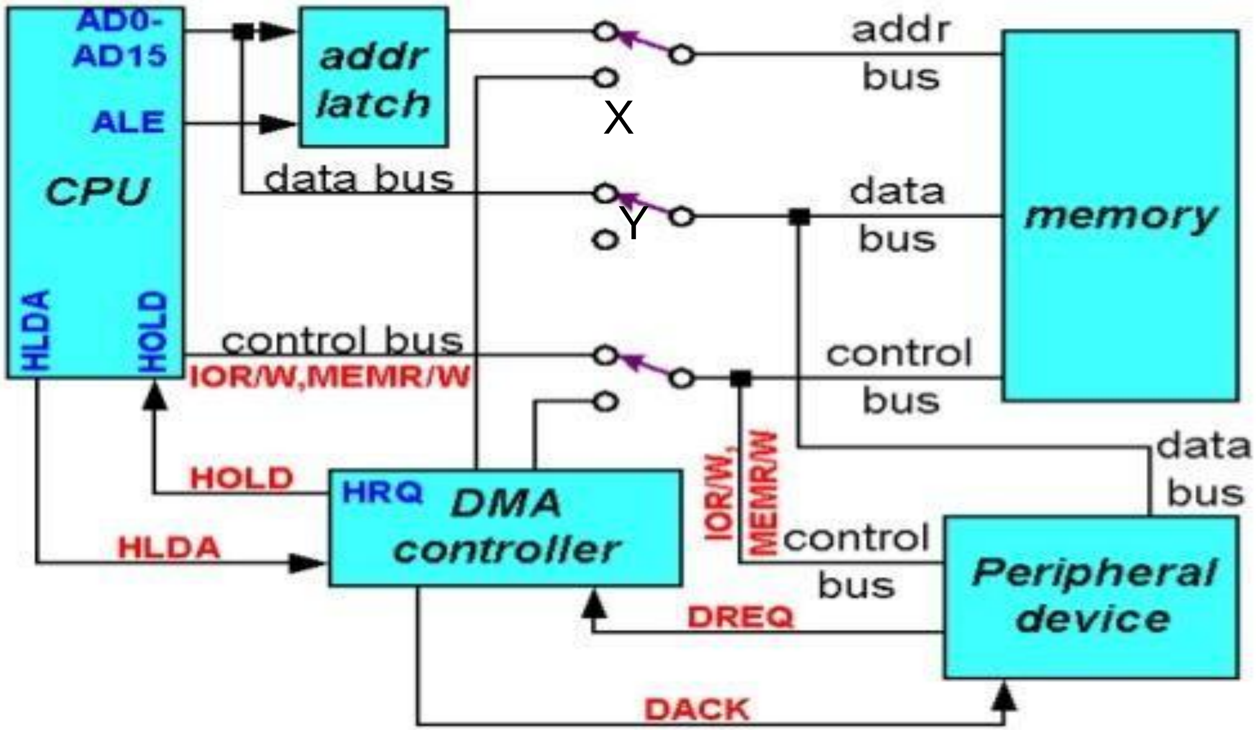
Operation of DMA Controller in Microprocessor

- In the **idle cycle** of the system, initially when the system gets on, then the processor has control over the system buses, as the switch are connected with the X position.
- In this position, the buses form the connection between main memory and peripherals through the processor. So, in this position, the processor performs the execution of the instruction.

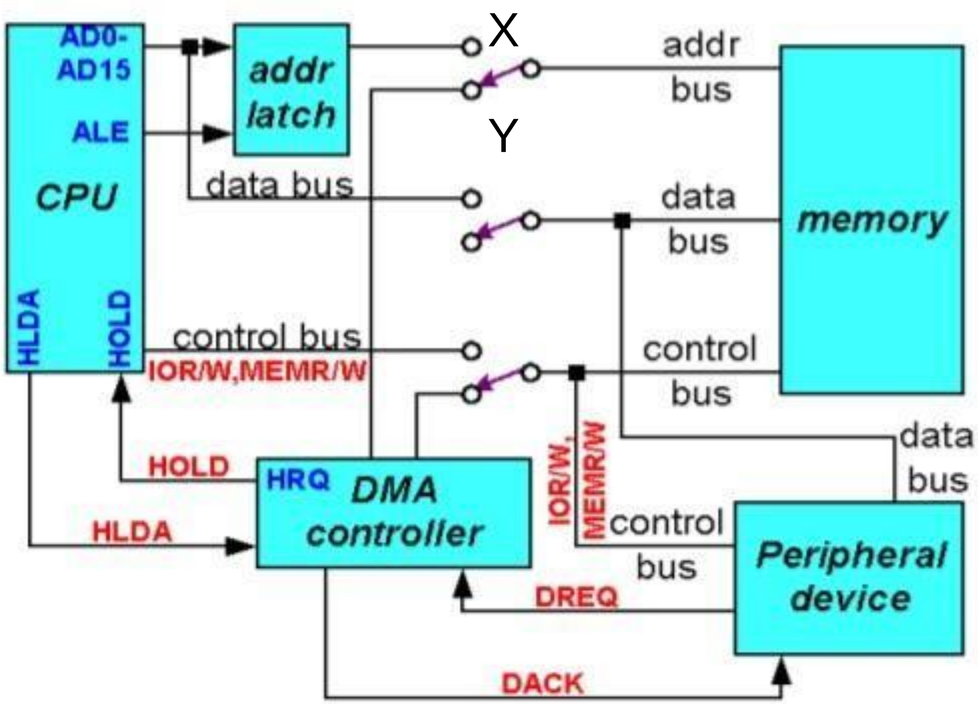
CPU having the control over the bus:



- When need arises to read the data from the disk. Then the microprocessor sends an instruction to the disk controller about the read operation of that particular data.
- On fetching the required data, the disk controller (peripheral device) sends DMA request, i.e., DRQ signal to the DMA controller. This DRQ signal shows that the device directly wants to transfer the data to the memory without disturbing the processor.

- on receiving the DRQ signal, HOLD request i.e., HRQ signal is sent by the DMA controller to the microprocessor.
- The HRQ signal shows the interest of the DMA controller to have access to system buses. So, on receiving HOLD request, the CPU tristates its buses in order to grant the control to the DMA controller.
- Once the processor frees the buses, then it sends the HLDA signal to the DMA controller. And on receiving HLDA signal, the control over the buses is given to the DMA controller as the switch position now changes from X to Y.
- So, gaining control over the buses, the **active cycle** of the DMA gets enabled. Thus now it sends the acknowledge signal DACK to the disk controller that shows that it is now ready for the transfer of data.

When DMA operates:



Basic Process of DMA-Min mode

- The **HOLD** and **HLDA** pins are used to receive and acknowledge the hold request respectively
- Normally the CPU has full control of the system bus.
- In a DMA operation, the peripheral takes over bus control temporarily.

A DMA controller interfaces with several peripherals that may request DMA.

- The controller decides the priority of simultaneous DMA requests communicates with the peripheral and the CPU, and provides memory addresses for data transfer.

- DMA controller commonly used with 8086 is the 8257/8237

programmable device.

- The 8257/8237 is a 4-channel device.

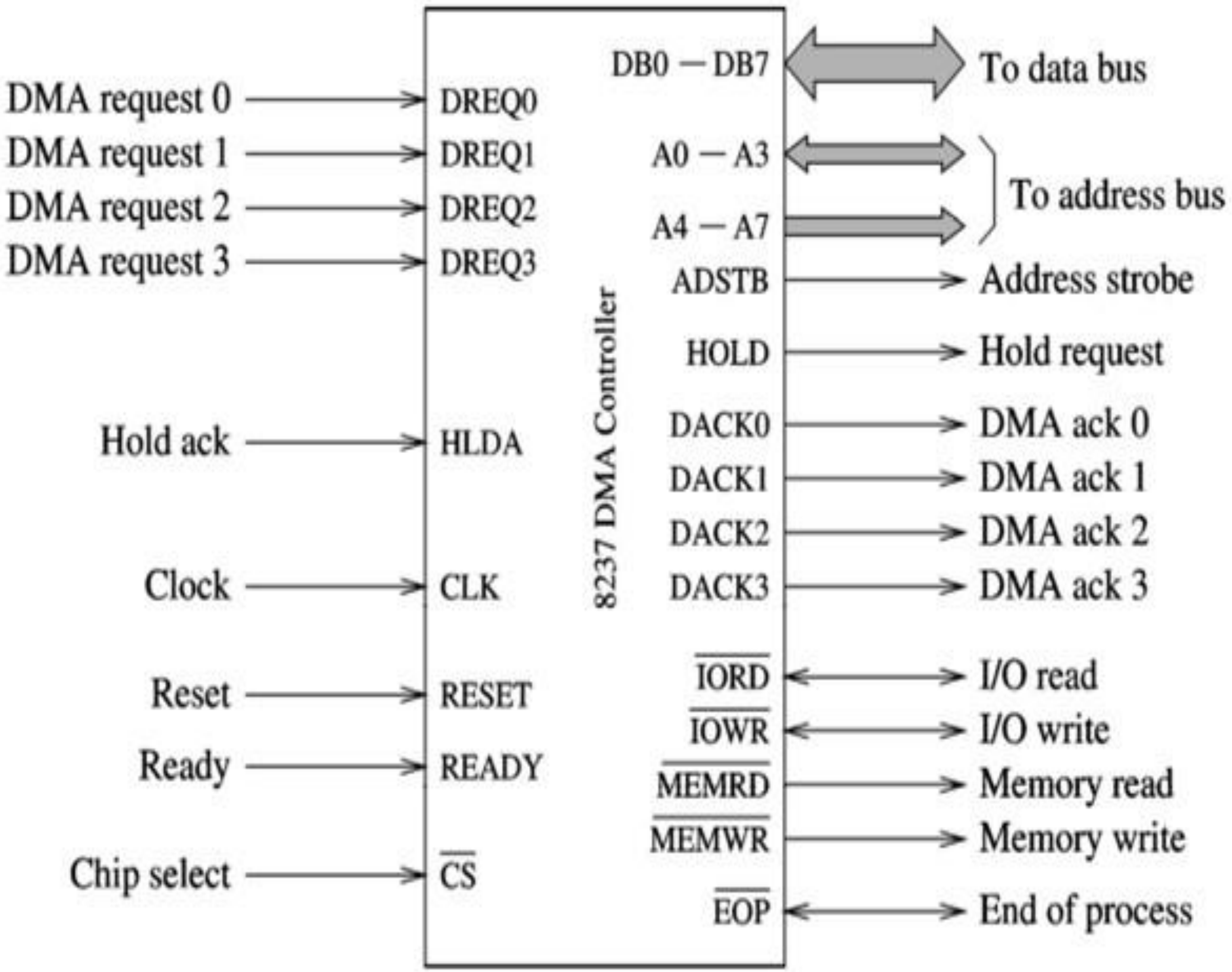
Each channel is dedicated to a specific peripheral device and capable of addressing 64 K bytes section of memory.

Features of 8237

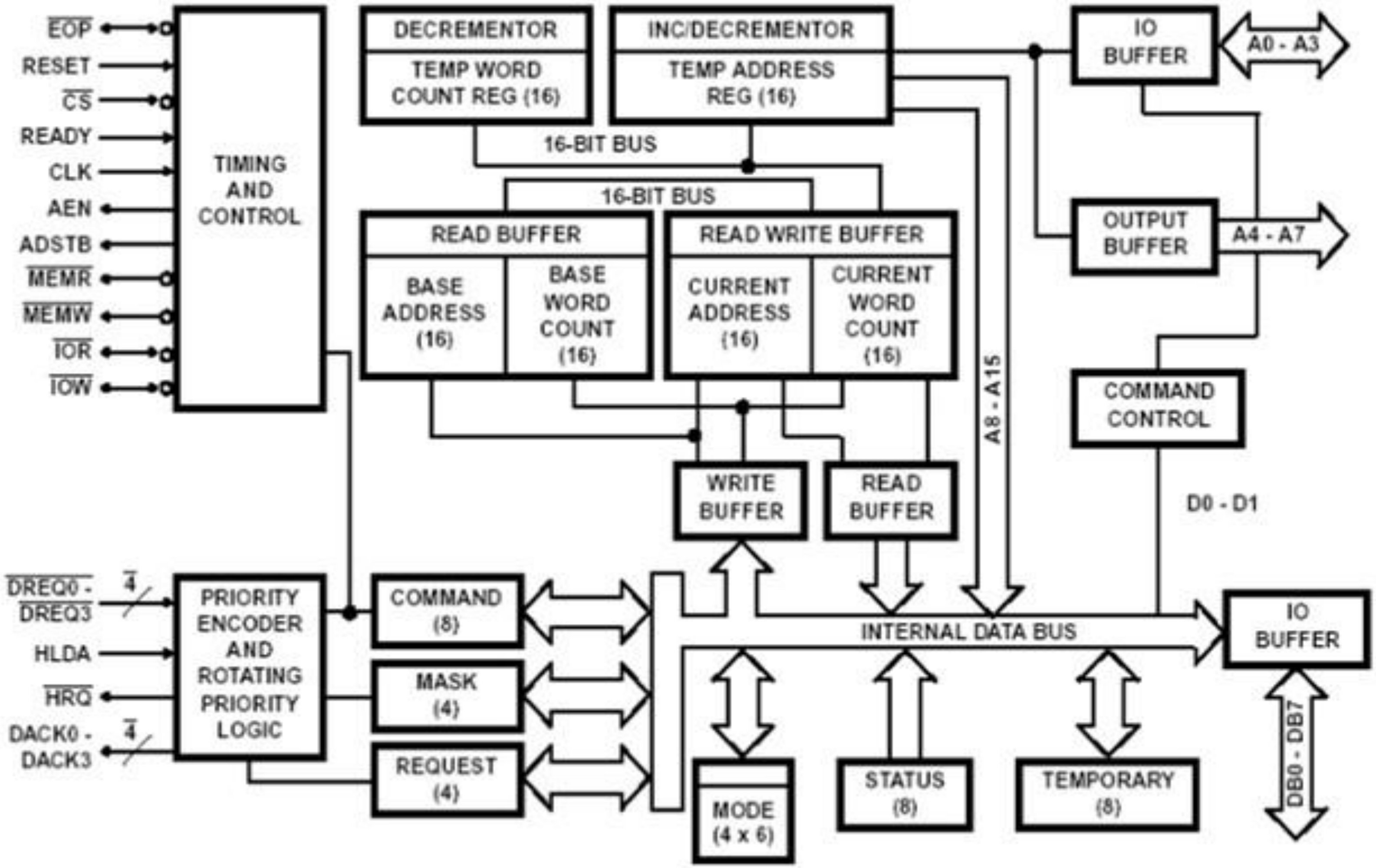
1. It provides various **modes of DMA.**
2. It provides on chip **4 independent channels**.
3. Number of channel can be increased by **cascading**.
4. Each channel can be used in **auto initialize** mode.
5. It can transfer data between memory to memory.
6. Address of memory is either i**ncremented or decremented.**
7. Clock frequency 3Mhz.
8. Data transfer rate 1.6 Mbps/sec.

9. Directly expendable to any no of channel by cascading.
10. It provides **EOP** line that is used for terminate DMA operation. This signal can be generated by external h/w.
11. DMA can be requested by setting an appropriate bit of request register.
12. Independent control for **DREQ and DACK**. These signal can be initialize by active high or low.
13. It provide compressed timing to improve throughput. it can compress the transfer time to two (2s)

8237 - DMA Controller



Block Diagram



8237 Pin Definitions

- CLOCK INPUT: Clock Input controls the internal operations of the 8237A and its rate of data transfers. The input may be driven at up to 5 MHz for the 8237A-5
- CHIP SELECT: Chip Select is an active low input used to select the 8237A as an I/O device during the Idle cycle. This allows CPU communication on the data bus

RESET: Reset is an active high input which clears the Command, Status, Request and Temporary registers. It also clears the first/ last flip/flop and sets the Mask register. Following a Reset the device is in the Idle cycle.

- READY: Ready is an input used to extend the memory read and write pulses from the 8237A to accommodate slow memories or I/O peripheral devices.
- HOLD ACKNOWLEDGE: The active high Hold Acknowledge from the CPU indicates that it has relinquished control of the system busses.
- MEMR’ **Memory read** is an output that causes memory to read data during a DMA read cycle.
- MEMW’ **Memory write** is an output that causes memory to write data during a DMA write cycle.

DB_0–DB_7

- **Data bus** pins are connected to the processor data bus connections and used during the programming of the DMA controller.

$\overline{IOR}$

- I/O READ: I/O Read is a bidirectional active low three-state line. In the Idle cycle, it is an input control signal used by the CPU to read the control registers. In the Active cycle, it is an output control signal used by the 8237A to access data from a peripheral during a DMA Write transfer.

A0-A3

The four least significant address lines are bidirectional three-state signals. In the Idle cycle they are inputs and are used by the CPU to address the register to be loaded or read. In the Active cycle they are outputs and provide the lower 4 bits of the output address

$\overline{\text{IOW}}$

I/O WRITE: I/O Write is a bidirectional active low three-state line. In the Idle cycle, it is an input control signal used by the CPU to load information into the 8237A. In the Active cycle, it is an output control signal used by the 8237A to load data to the peripheral during a DMA Read transfer.

$DREQ_0$–$DREQ_3$

DMA REQUEST: The DMA Request lines are individual asynchronous channel request inputs used by peripheral circuits to obtain DMA service. In fixed Priority, DREQ0 has the highest priority and DREQ3 has the lowest priority. A request is generated by activating the DREQ line of a channel. DACK will acknowledge the recognition of DREQ signal. Polarity of DREQ is programmable. Reset initializes these lines to active high. DREQ must be maintained until the corresponding DACK goes active.

EOP: END OF PROCESS:

End of Process is an active low bidirectional signal. Information concerning the completion of DMA services is available at the bidirectional EOP pin. The 8237A allows an external signal to terminate an active DMA service. This is accomplished by pulling the EOP input low with an external EOP signal. The 8237A also generates a pulse when the terminal count (TC) for any channel is reached. This generates an EOP signal which is output through the EOP line. The reception of EOP, either internal or external, will cause the 8237A to terminate the service, reset the request, and, if Autoinitialize is enabled, to write the base registers to the current registers of that channel.

HRQ

- **Hold request** is an output that connects to the HOLD input of the microprocessor in order to request a DMA transfer.

DACK0-DACK3

- **DMA channel acknowledge** outputs acknowledge a channel DMA request.
- These outputs are programmable as either active-high or active-low signals.
  - DACK outputs are often used to select the DMA- controlled I/O device during the DMA transfer.

AEN

- **Address enable** signal enables the DMA address latch connected to the DB_7–DB_0 pins on the 8237.
  - – also used to disable any buffers in the system connected to the microprocessor

ADSTB

Address strobe functions as ALE, except it is used by the DMA controller to latch address bits A_{15}–A_8 during the DMA transfer

ADDRESS A4 –A7: The four most significant address lines are three-state outputs and provide 4 bits of address. These lines are enabled only during the DMA service.

8237 Internal Registers

CAR

- The **current address register** holds a 16-bit memory address used for the DMA transfer.
  - each channel has its own current address register for this purpose
- When a byte of data is transferred during a DMA operation, CAR is either incremented or decremented.
  - depending on how it is programmed

8237 Internal Registers

CWCR

- The **current word count register** programs a channel for the number of bytes (up to 64K) transferred during a DMA action.
- The number loaded into this register is one less than the number of bytes transferred.
  - for example, if a 10 is loaded to CWCR, then 11 bytes are transferred during the DMA action

8237 Internal Registers

BA and BWC

- The **base address** (BA) and **base word count** (BWC) registers are used when auto-initialization is selected for a channel.
- In auto-initialization mode, these registers are used to reload the CAR and CWCR after the DMA action is completed.
  - allows the same count and address to be used to transfer data from the same memory area

Single Transfer Mode :In Single Transfer mode the device is programmed to make one transfer only. The word count will be decremented and the address decremented or incremented following each transfer. When the word count ‘‘rolls over’’ from zero to FFFFH, a Terminal Count (TC) will cause an Autoinitialize if the channel has been programmed to do so

Block Transfer Mode-In Block Transfer mode the device is activated by DREQ to continue making transfers during the service until a TC, caused by word count going to FFFFH, or an external End of Process (EOP) is encountered. DREQ need only be held active until DACK becomes active. Again, an Autoinitialization will occur at the end of the service if the channel has been programmed for it.

Demand Transfer Mode-In Demand Transfer mode the device is programmed to continue making transfers until a TC or external EOP is encountered or until DREQ goes inactive. Thus transfers may continue until the I/O device has exhausted its data capacity.

Cascade Mode-This mode is used to cascade more than one 8237A together for simple system expansion. The HRQ and HLDA signals from the additional 8237A are connected to the DREQ and DACK signals of a channel of the initial 8237A.

each of the three active transfer modes can perform three different types of transfers. These are read, write, and verify.

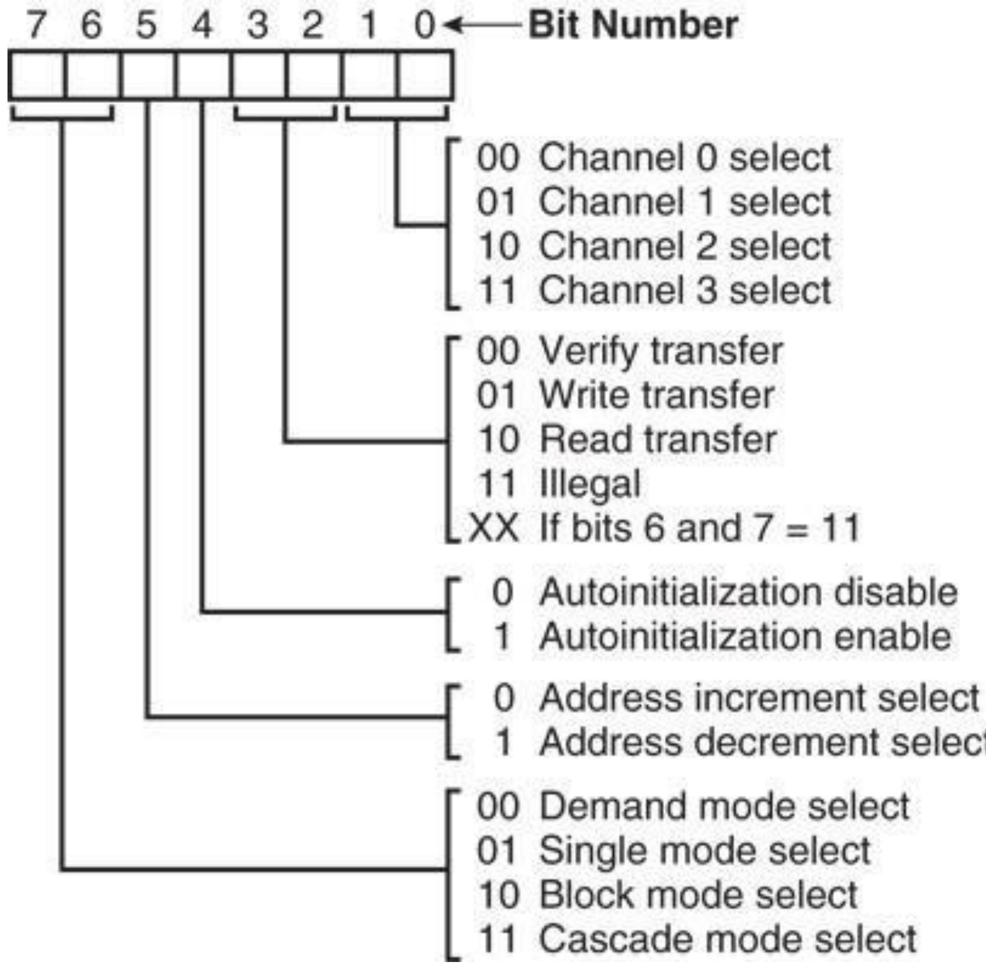
- Write transfers move data from an I/O device to the memory by activating MEMW and IOR.
- Read transfers move data from memory to an I/O device by activat- ing MEMR and IOW.
- Verify transfers are pseudotransfers. The 8237A operates as in read or write transfers generating addresses, and responding to EOP, etc. However, the memory and I/O control lines all remain inac- tive. The ready input is ignored in verify mode.

8237 Internal Registers

MR

- The **mode register** programs the mode of operation for a channel.
- Each channel has its own mode register as selected by bit positions 1 and 0.
  - remaining bits of the mode register select operation, auto-initialization, increment/decrement, and mode for the channel

Figure 13–5 8237A-5 mode register. (Courtesy of Intel Corporation.)



Enabled (Address Hold Disabled): If the "Channel 0 Address Hold Disable" bit is set, then the memory address for data transfers on Channel 0 will not automatically increment. This can be useful for certain types of memory-to-memory transfers where you want to transfer data from a fixed source address to a fixed destination address.

Disabled (Address Hold Enabled): If the "Channel 0 Address Hold Disable" bit is cleared (not set), then the memory address for data transfers on Channel 0 will automatically increment after each data transfer. This is more typical behavior for other DMA channels used for block transfers.

Controller Enable: When the DMA controller is in the "Controller Enable" state, it is actively functioning and ready to perform data transfers. In this state, it can respond to requests from I/O devices or software to initiate DMA transfers. When enabled, the DMA controller will execute the data transfer instructions as configured for each of its channels. The enable state allows the DMA controller to operate and perform data transfers as needed.

Controller Disable: Conversely, when the DMA controller is in the "Controller Disable" state, it is effectively paused or turned off. In this state, the DMA controller will not respond to any requests for data transfers, and ongoing transfers may be halted. This can be useful in situations where you want to temporarily prevent the DMA controller from accessing memory or managing data transfers. Disabling the controller allows the CPU to have full control over memory and I/O operations.

Priority of 8237

The 8237A has two types of priority encoding available as software selectable options.
The first is Fixed Priority which fixes the channels in priority order based upon the descending value of their number. The channel with the lowest priority is 3 followed by 2, 1 and the highest priority channel, 0. After the recognition of any one channel for service, the other channels are prevented from interfering with that service until it is completed.

After completion of a service, HRQ will go inactive and the 8237A will wait for HLDA to go low before activating HRQ to service another channel.

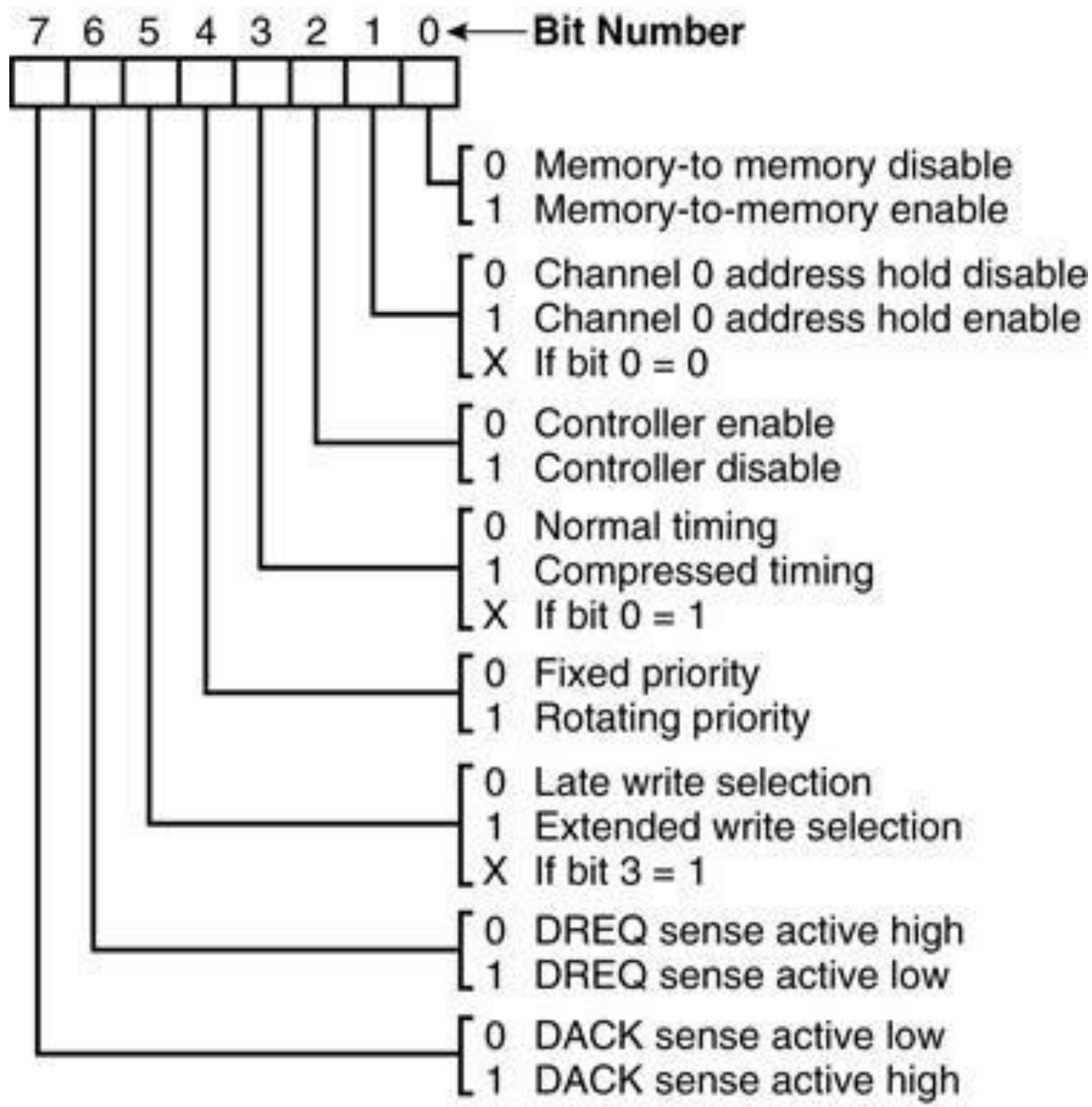
The second scheme is Rotating Priority. The last channel to get service becomes the lowest priority channel with the others rotating accordingly. With Rotating Priority in a single chip DMA system, any device requesting service is guaranteed to be recognized after no more than three higher priority services have occurred. This prevents any one channel from monopolizing the system

8237 Internal Registers

CR

- The **command register** programs the operation of the 8237 DMA controller.
- The register uses bit position 0 to select the memory-to-memory DMA transfer mode.
  - memory-to-memory DMA transfers use DMA channel 0 to hold the source address
  - DMA channel 1 holds the destination address
- Similar to operation of a MOVSB instruction.

Figure 13–4 8237A-5 command register. (Courtesy of Intel Corporation.)

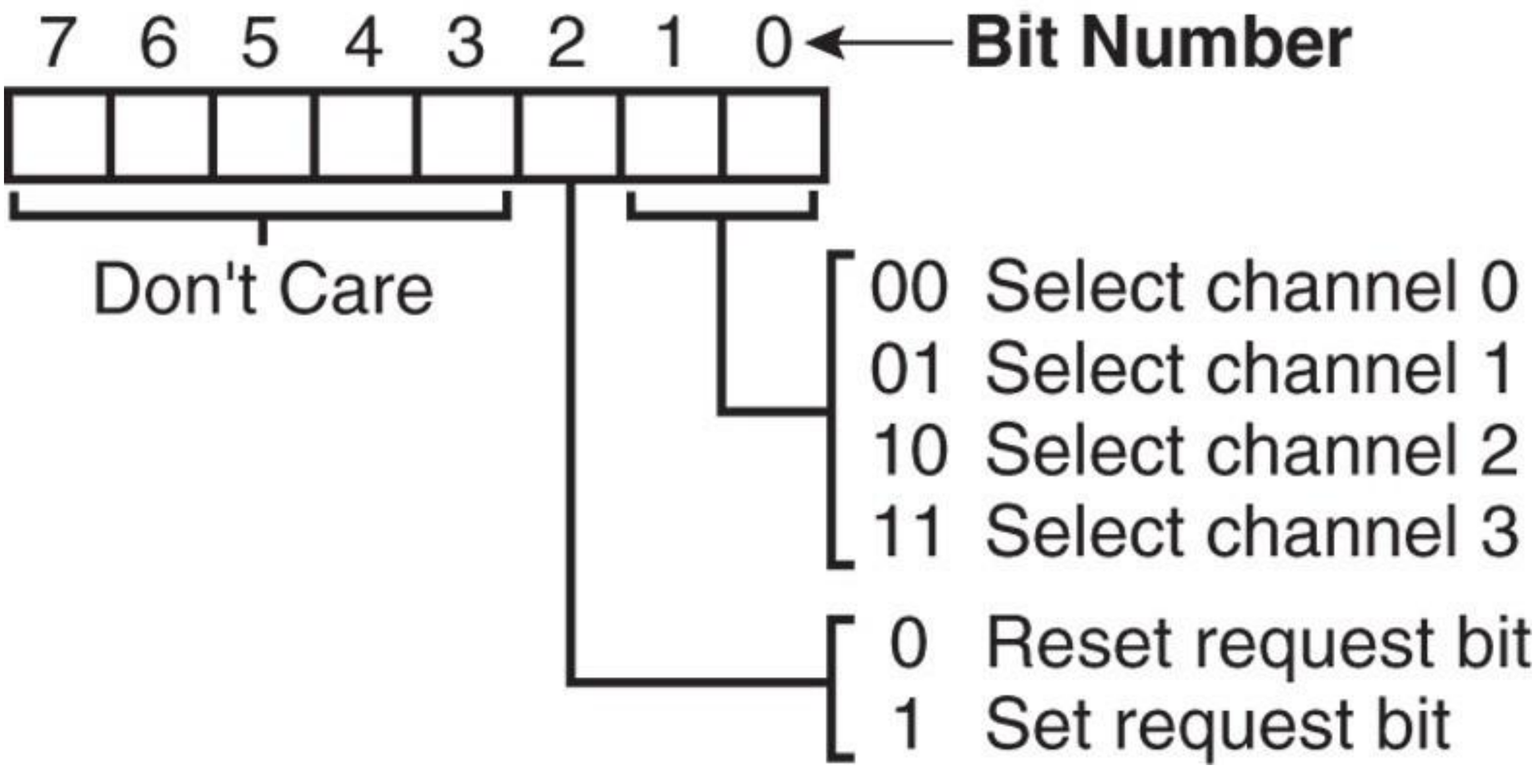


8237 Internal Registers

BR

- The **bus request register** is used to request a DMA transfer via software.
  - very useful in memory-to-memory transfers, where an external signal is not available to begin the DMA transfer

Figure 13–6 8237A-5 request register. (Courtesy of Intel Corporation.)

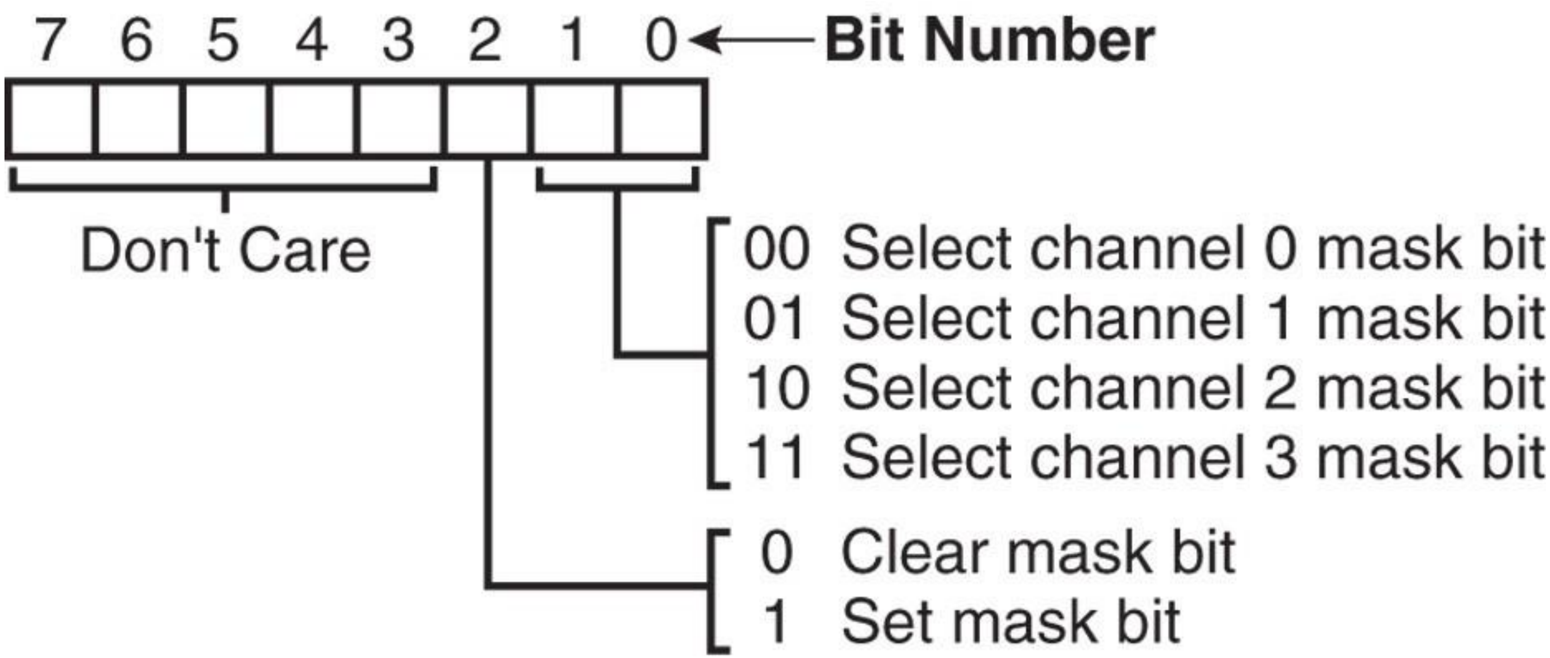


8237 Internal Registers

MRSR

- The **mask register set/reset** sets or clears the channel mask.
  - if the mask is set, the channel is disabled
  - the RESET signal sets all channel masks to disable them

Figure 13–7 8237A-5 mask register set/reset mode. (Courtesy of Intel Corporation.)

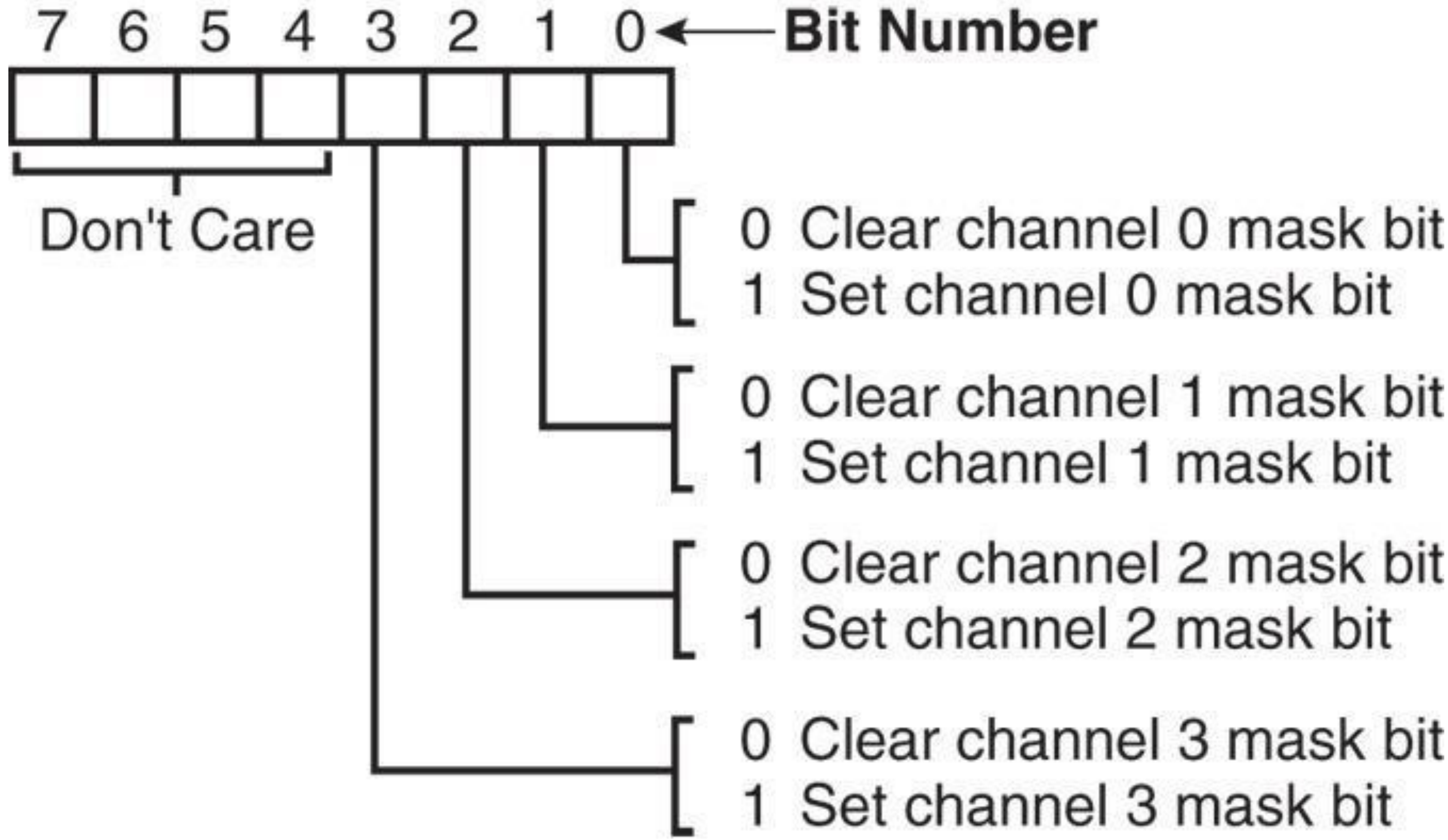


8237 Internal Registers

MSR

- The **mask register** clears or sets all of the masks with one command instead of individual channels, as with the MRSR.

Figure 13–8 8237A-5 mask register. (Courtesy of Intel Corporation.)



8237 Internal Registers

SR

- The **status register** shows status of each DMA channel. The TC bits indicate if the channel has reached its terminal count (transferred all its bytes).
- When the terminal count is reached, the DMA transfer is terminated for most modes of operation.
  - the request bits indicate whether the DREQ input for a given channel is active

Figure 13–9 8237A-5 status register. (Courtesy of Intel Corporation.)

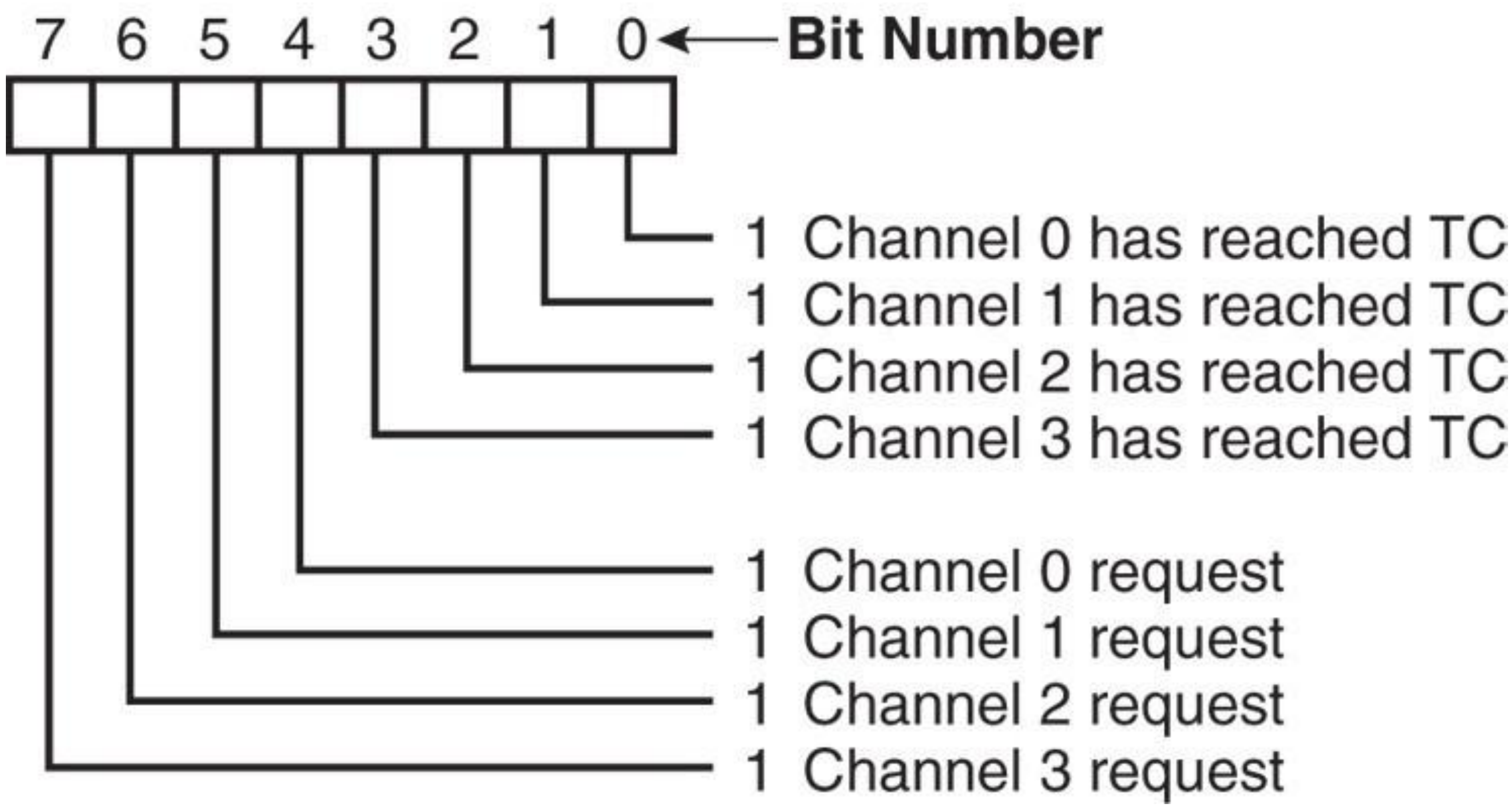


Figure 13–10 8237A-5 command and control port assignments. (Courtesy of Intel Corporation.)

Signals						Operation
A3	A2	A1	A0	$\overline{IOR}$	$\overline{IOW}$	
1	0	0	0	0	1	Read Status Register
1	0	0	0	1	0	Write Command Register
1	0	0	1	0	1	Illegal
1	0	0	1	1	0	Write Request Register
1	0	1	0	0	1	Illegal
1	0	1	0	1	0	Write Single Mask Register Bit
1	0	1	1	0	1	Illegal
1	0	1	1	1	0	Write Mode Register
1	1	0	0	0	1	Illegal
1	1	0	0	1	0	Clear Byte Pointer Flip/Flop
1	1	0	1	0	1	Read Temporary Register
1	1	0	1	1	0	Master Clear
1	1	1	0	0	1	Illegal
1	1	1	0	1	0	Clear Mask Register
1	1	1	1	0	1	Illegal
1	1	1	1	1	0	Write All Mask Register Bits

8237 Software Commands

Master clear

- Acts exactly the same as the RESET signal to the 8237.
  - as with the RESET signal, this command disables all channels

Clear mask register

- Enables all four DMA channels.

8237 Software Commands

Clear the first/last flip-flop

- Clears the first/last (F/L) flip-flop within 8237.
- The F/L flip-flop selects which byte (low or high order) is read/written in the current address and current count registers.
  - if F/L = 0, the low-order byte is selected
  - if F/L = 1, the high-order byte is selected
- Any read or write to the address or count register automatically toggles the F/L flip-flop.

Programming the Address and Count Registers

- Figure shows I/O port locations for programming the count and address registers for each channel.
- The state of the F/L flip-flop determines whether the LSB or MSB is programmed.
  - if the state is unknown, count and address could be programmed incorrectly
- It is important to disable the DMA channel before address and count are programmed.

Figure 8237A-5 DMA channel I/O port addresses. (Courtesy of Intel Corporation.)

Channel	Register	Operation	Signals							Internal Flip-Flop	Data Bus DB0-DB7
			$\overline{CS}$	$\overline{IOR}$	$\overline{IOW}$	A3	A2	A1	A0		
0	Base and Current Address	Write	0 0	1 1	0 0	0 0	0 0	0 0	0 0	0 1	A0-A7 A8-A15
	Current Address	Read	0 0	0 0	1 1	0 0	0 0	0 0	0 0	0 1	A0-A7 A8-A15
	Base and Current Word Count	Write	0 0	1 1	0 0	0 0	0 0	0 0	1 1	0 1	W0-W7 W8-W15
	Current Word Count	Read	0 0	0 0	1 1	0 0	0 0	0 0	1 1	0 1	W0-W7 W8-W15
1	Base and Current Address	Write	0 0	1 1	0 0	0 0	0 0	1 1	0 0	0 1	A0-A7 A8-A15
	Current Address	Read	0 0	0 0	1 1	0 0	0 0	1 1	0 0	0 1	A0-A7 A8-A15
	Base and Current Word Count	Write	0 0	1 1	0 0	0 0	0 0	1 1	1 1	0 1	W0-W7 W8-W15
	Current Word Count	Read	0 0	0 0	1 1	0 0	0 0	1 1	1 1	0 1	W0-W7 W8-W15
2	Base and Current Address	Write	0 0	1 1	0 0	0 0	1 1	0 0	0 0	0 1	A0-A7 A8-A15
	Current Address	Read	0 0	0 0	1 1	0 0	1 1	0 0	0 0	0 1	A0-A7 A8-A15
	Base and Current Word Count	Write	0 0	1 1	0 0	0 0	1 1	0 0	1 1	0 1	W0-W7 W8-W15
	Current Word Count	Read	0 0	0 0	1 1	0 0	1 1	0 0	1 1	0 1	W0-W7 W8-W15
3	Base and Current Address	Write	0 0	1 1	0 0	0 0	1 1	1 1	0 0	0 1	A0-A7 A8-A15
	Current Address	Read	0 0	0 0	1 1	0 0	1 1	1 1	0 0	0 1	A0-A7 A8-A15
	Base and Current Word Count	Write	0 0	1 1	0 0	0 0	1 1	1 1	1 1	0 1	W0-W7 W8-W15
	Current Word Count	Read	0 0	0 0	1 1	0 0	1 1	1 1	1 1	0 1	W0-W7 W8-W15

- Four steps are required to program the 8237:
  - (1) The F/L flip-flop is cleared using a clear F/L command
  - (2) the channel is disabled
  - (3) LSB & MSB of the address are programmed
  - (4) LSB & MSB of the count are programmed
- Once these four operations are performed, the channel is programmed and ready to use.
  - additional programming is required to select the mode of operation before the channel is enabled and started

Example

- Design the 8237 decoding circuit and the 8237 address line connections so that the 8237 is in the address range 70h-7Fh
- Write a program that starts a block memory-to-memory DMA transfer from memory locations 10000H-13FFFH to 14000H-17FFFH using channel 0 as source and channel 1 as destination.

8237 Programming Example

```
CLEAR_FF        EQU 7CH ;F/L CLEAR VALUE
CH0_A           EQU 70H ;CHANNEL 0 ADDRESS
CH1_A           EQU 72H ;CHANNEL 1 ADDRESS
CH1_C           EQU 73H ;CHANNEL 1 COUNT
MODE            EQU 7BH ;MODE
CR              EQU 78H ;COMMAND REGISTER
MASKS           EQU 7FH ;MASKS
REQ             EQU 79H ;REQUEST REGISTER
STATUS          EQU 78H ;STATUS REGISTER
   ;ES = segment of source and destination
   ;SI = source address
   ;DI = destination address
   ;CX = count
DMA PROC FAR
   MOV AL, 0
   OUT CLEAR_FF, AL    ;CLEAR F/L FF
   MOV AX, ES          ;PROGRAM SOURCE ADDRESS
   SHL AX, 4           ;SHIFT LEFT SEGMENT
   ADD AX, SI          ;ADD SOURCE OFFSET
   OUT CH0_A, AL       ;CHANNEL 0 ADDRESS PROGRAMMING LSB FIRST
   MOV AL, AH          ;ONLY AL ALLOWED IN IN/OUT INSTRUCTIONS
   OUT CH0_A, AL       ;CHANNEL 0 ADDRESS PROGRAMMING MSB LAST
```

EXAMPLE (CONTINUED)

```
MOV AX, ES      ;PROGRAM DESTINATION ADDRESS
SHL AX, 4       ;SHIFT LEFT SEGMENT
ADD AX, DI      ;ADD DESTINATION OFFSET
OUT CH1_A, AL   ;CHANNEL 1 ADDRESS PROGRAMMING LSB FIRST
MOV AL, AH      ;ONLY AL ALLOWED IN IN/OUT INSTRUCTIONS
OUT CH1_A, AL   ;CHANNEL 1 ADDRESS PROGRAMMING MSB FIRST
MOV AX, CX      ;PROGRAM COUNT
DEC AX          ;ADJUST COUNT
OUT CH1_C, AL   ;MOVE TO CHANNEL 1 COUNT
MOV AL, AH
OUT CH1_C, AL
MOV AL, 88H     ;PROGRAM MODE
OUT MODE, AL
MOV AL,3        ;MEMORY-TO-MEMORY TRANSFER
OUT CR, AL
MOV AL, 0EH     ;UNMASK CHANNEL 0
OUT MASKS, AL
MOV AL, 4       ;START DMA TRANSFER BY SETTING REQUEST BIT FOR
                       CHANNEL 0
OUT REQ, AL
```